Year 13 Further Maths - Matrices

Similar questions - Powers, proof by induction and 3D transformations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

(a) Find $\mathbf{A}^2$, $\mathbf{A}^3$ and $\mathbf{A}^4$ and make a conjecture for $\mathbf{A}^n$, for $n \geqslant 1$. **[3]**

(b) Prove your conjecture by induction. **[5]**

2

(a) $\mathbf{R} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Describe the transformation represented by $\mathbf{R}$ and find $\mathbf{R}^4$. **[3]**

(b) Hence find $\mathbf{R}^{423}$, justifying your answer. **[3]**

(c) $\mathbf{S} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Show that $\mathbf{S}^2 = \mathbf{I}$ and describe the transformation represented by $\mathbf{S}$. **[4]**

(d) The matrix $\mathbf{B}$ represents a reflection in the plane $y = 0$. Write down $\mathbf{B}$ and find the image of the point $(1, 2, 3)$ under $\mathbf{B}$. **[2]**

Question 1

(a) $\mathbf{A}^2 = \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}$, $\mathbf{A}^3 = \begin{pmatrix} 1 & 7 \\ 0 & 8 \end{pmatrix}$, $\mathbf{A}^4 = \begin{pmatrix} 1 & 15 \\ 0 & 16 \end{pmatrix}$. Conjecture: $\mathbf{A}^n = \begin{pmatrix} 1 & 2^n - 1 \\ 0 & 2^n \end{pmatrix}$

(b) Base case $n = 1$: $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = \mathbf{A}$, true. Assume true for $n = k$. Then $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \begin{pmatrix} 1 & 2^k - 1 \\ 0 & 2^k \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 + 2(2^k - 1) \\ 0 & 2^{k+1} \end{pmatrix} = \begin{pmatrix} 1 & 2^{k+1} - 1 \\ 0 & 2^{k+1} \end{pmatrix}$, which is the formula for $n = k + 1$. True for $n = 1$ and true for $n = k \Rightarrow$ true for $n = k + 1$, so true for all positive integers n

Question 2

(a) $\mathbf{R}$ is a rotation of 90° anticlockwise about the z-axis (looking down from positive z). $\mathbf{R}^4 = \mathbf{I}$

(b) $423 = 4 \times 105 + 3$, so $\mathbf{R}^{423} = (\mathbf{R}^4)^{105}\mathbf{R}^3 = \mathbf{R}^3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (rotation of 270°, i.e. 90° clockwise)

(c) $\mathbf{S}^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. $\det\mathbf{S} = +1$ and $\mathbf{S}$ fixes $(1, 1, 0)$ and reverses $(1, -1, 0)$ and $(0, 0, 1)$: a rotation of 180° about the line $x = y$, $z = 0$

(d) $\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and the image of $(1, 2, 3)$ is $(1, -2, 3)$